

PHASE 1: PRIVACY & INGESTION GUARDRAILS

Raw Domain Corpus Ingestion
JSONL / Unstructured Text Records

Regex + NER Redaction Engine
SSN, Email, Phone, IP, CC, API Keys

Volatile RAM Buffer & Shredder
Shared RAM (/dev/shm) | 3-Pass DoD Wipe

✓ Masked Tokens

PHASE 2: ALTERNATING DP-LORA TRAINING

Frozen Backbone + LoRA Adapters
Base LLM W_0 (Frozen) | Adapters $\Delta W = B \cdot A$ ($r=8$)

LA-LoRA Alternating Updates
Step 2k-1: Update A (Freeze B) | Step 2k: Update B

DP-SGD & RDP Privacy Accountant
Gaussian Noise $N(0, \sigma^2 C^2 I)$ | (ϵ, δ) -DP ($\epsilon < 8.0$)

✓ LoRA Tensors

PHASE 3: HARDWARE BINDING & CRYPTO

Hardware Fingerprint Probe (F)
CPU Microcode || OS UUID || Disk Serial

HKDF Key & AES-256-GCM Stream
 $K_{\text{master}} = \text{HKDF}(F, s)$ | Up to 4,602 MB/s Enc

RSA-2048-PSS Signed Package
Header Signature $\sigma_H \rightarrow$ Artifact: adapter.seclora

✓ adapter.seclora

PHASE 4: RUNTIME ENCLAVE DEPLOYMENT

8-Gate Verification Engine
Header, RSA Sig, HWID, Salt, HKDF, GCM Tag, Digest

Docker CGroup Execution Enclave
Isolated VRAM Allocation | Zero Plaintext at Rest

Secure Zero-Latency Inference
1.33 ms Total Latency | 100% STRIDE Pass